

Waste Audit Report (6-7 Weeks)

COURSE CODE: SE-118

COURSE TITLE:

Civics & Community Engagements

REPORT TITLE:

The Eco-Diplomacy Report (6-7 Weeks)

REPORT ASSESSMENT TABLE:

Obtained Marks	Total Marks

Submitted To: Dr. Ghousia Saeed.

Submitted By: Muhammad Danial &
Syed Zameer Bukhari.

Registration – No: 24ABSWE0021 &
24ABSWE0032.

Semester: 4th

Date: 17-07-2026.

Dept. of Software Engineering

UET Peshawar, Abbottabad Campus.



Instructor Sign : _____



Department of Software Engineering
University of Engineering and Technology, Peshawar, Abbottabad Campus

BSI-121/ SE-118: Civics and Community Engagement

4th Semester - Spring 2026

Eco-Diplomacy: A Student-Led Circular Economy Solution for Campus and Community Waste
Management

Project Group & Site:

21	24ABSWE0021	MUHAMMAD DANIAL	S10	Old College Road
32	24ABSWE0032	SYED ZAMIR BUKHARI		

Phase-by-Phase Alignment (6-7 Weeks)

Phase 1: Situational Audit (Week 1)

Activity: Site survey, observe current waste management practices, identify types of waste,
Recognize environmental and social impacts and stakeholder identification.

Activity

- Site survey of campus/community.
- Observe existing waste management practices.
- Identify waste types and quantities.
- Identify stakeholders.
- Recognize environmental and social impacts.

Introduction

A situational audit was conducted by (**Muhammad Danial & Syed Zameer Bukhari**) at Old College Road to examine the current waste management condition in the community. The purpose of this activity was to observe waste disposal practices, identify the major waste types, recognize responsible stakeholders and understand the environmental and social impacts caused by unmanaged waste.

Survey Evidence (Pictures)

The following pictures and video provide visual evidence of the waste accumulation, polluted drainage water, and improper disposal practices observed during the site survey. These visuals support the findings recorded during the situational audit.



Open dumping of mixed waste near polluted drainage water at Old College Road.



Accumulated plastic and organic waste affecting community cleanliness.



Contaminated water and unmanaged garbage creating public health risks.

Activities Performed

1. Site Survey of Community

A physical survey of the Old College Road area was conducted to observe waste disposal locations and surrounding environmental conditions. Large amounts of garbage were found accumulated near open drainage water.

2. Observation of Existing Waste Management Practices

It was observed that waste is being dumped openly without proper collection or segregation. No effective waste disposal system or cleanliness monitoring was visible in the surveyed area.

3. Identification of Waste Types and Quantities

Different types of waste were identified during the survey, including:

- Plastic bags and wrappers.
- Food and organic waste.
- Poultry/chicken waste.
- Straw and market waste.
- Household garbage.

The quantity of waste was significant, especially near the drainage area where waste had accumulated continuously.



4. Identification of Stakeholders

The following stakeholders were identified:

- Local residents.
- Vegetable vendors and shoppers.
- Chicken shop owners.
- Municipal waste management authorities.
- Nearby community members affected by pollution.

5. Environmental and Social Impacts

The unmanaged waste caused visible water pollution and unpleasant surroundings. Poultry waste and garbage mixed with stagnant water may increase health risks and spread diseases. The area also produces foul smell, damages environmental cleanliness, and negatively affects the local community.

6. Conclusion

The situational audit highlighted serious waste management problems at Old College Road. Immediate attention, community awareness and proper waste management practices are necessary to improve environmental conditions and public health in the area.

Phase 3: Meeting Report (Week 3)

Activity: Presenting solutions to stakeholders, negotiating, and finalising the plan.

Activity

- Present proposed waste management solutions to stakeholders.
- Engage community representatives and campus authorities.
- Conduct discussion, negotiation, and feedback sessions.
- Modify solutions based on stakeholder input.
- Finalize an agreed implementation plan.

1. Introduction

The Eco-Diplomacy Meeting was conducted by (**Muhammad Danial & Syed Zameer Bukhari**) at Old College Road to engage key stakeholders to present, discuss, and refine the proposed waste management strategies for the university campus/surrounding community. The main objective was to ensure that the proposed interventions were practical, socially acceptable, environmentally sustainable and collectively agreed upon by all relevant stakeholders.

This phase emphasizes participatory decision-making, where solutions were not imposed but developed through dialogue, negotiation, and feedback. The process aligned with the principles of the circular economy, promoting waste reduction, resource efficiency, and regeneration of natural systems through the 5R framework (Refuse, Reduce, Reuse, Recycle, Rot).

2. Objectives of Phase 3

- To present waste management strategies to stakeholders in an accessible manner.



- To engage community representatives, students, and campus authorities.
- Conducting interactive discussions and collecting feedback.
- To negotiate and modify proposed solutions based on stakeholder input.
- To finalize an implementable and agreed waste management plan.

3. Stakeholders Involved

Stakeholder Group	Role in Project
Vegetable Sellers	Segregation of organic waste.
Dairy Farm Owner	Collection and utilization of organic waste as animal feed.
Mobile Shopkeepers	Proper handling of packaging waste.
Community Members	Participation in cleanliness activities.
Student Team	Awareness, coordination and monitoring.

Table 1: Stakeholders and their Roles.

4. Method of Presentation and Engagement

Instead of formal presentations, the solutions were shared through participatory and community-friendly methods. Since many stakeholders were more comfortable with informal communication, emphasis was placed on dialogue-based engagement.

Methods used included:

- 📍 Face-to-face meeting.
- 📍 Informal discussion circles.
- 📍 Verbal explanations in simple and local language.
- 📍 Open negotiations.

5. Meeting Evidence (Pictures)

The following pictures and video provide visual evidence of the Eco-Diplomacy Meeting.

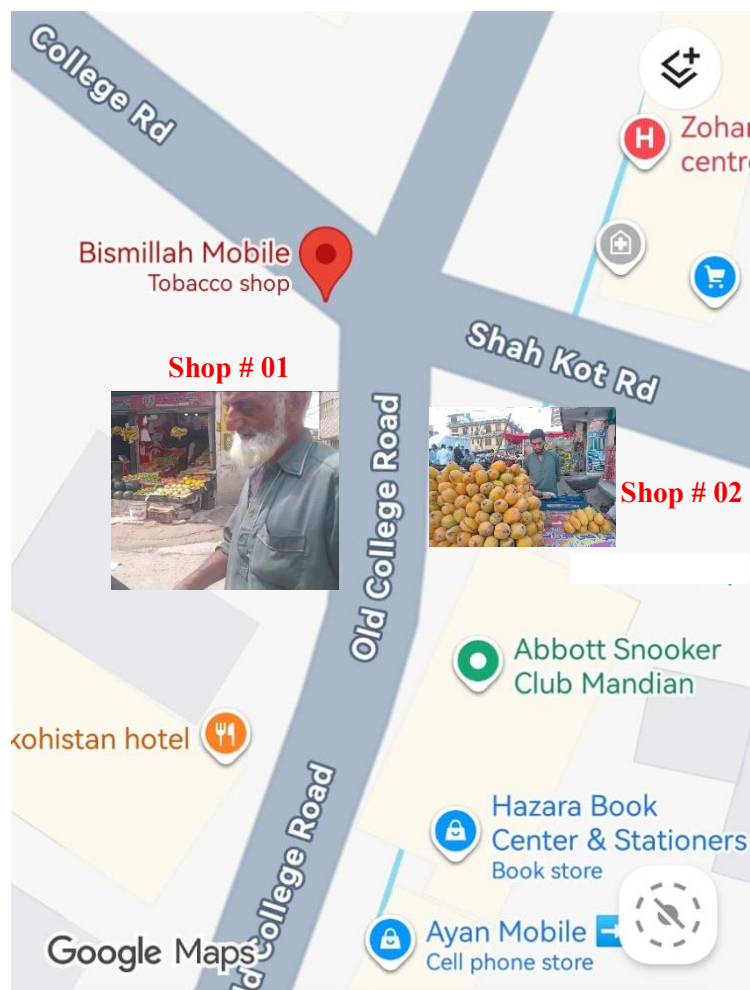
Stakeholder # 01: Vegetable Sellers (Organic Waste Segregator)



Pic 01: Negotiating with the Vegetable Shopkeepers.



Pic 02: Showing Segregated Wooden Fruit Baskets for the sake of burning at home as firewood..



Pic 03: Showing Google Map Location of 1st Stakeholder “Vegetable Shops”.



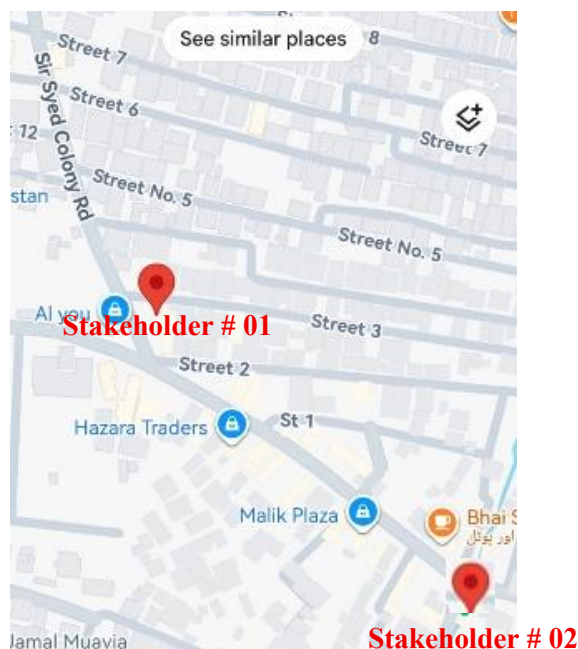
Stakeholder # 02: Dairy Milk Shop (Organic Waste Collector)



Pic 04: Showing Dairy Milk Shop Photos after meeting.



Pic 05: Showing Dairy Milk Shop (Stakeholder # 02) on Google Map.



Pic 06: Showing connection b/w two Stakeholders.



6. Key Solutions Presented (5R + Circular Economy Framework)

The proposed waste management system was designed under the 5R principles integrated with the economy approach, aiming to eliminate the concept of waste and treat materials as resources.

6.1. Refuse – Prevent Waste Generation

Stakeholders were encouraged to avoid unnecessary plastic bags and disposable packaging. Shopkeepers were advised to minimize excess packaging materials whenever possible.

Stakeholders discussed how this refusal practices could gradually reduce waste entering the management system.

6.2. Reduce – Minimize Resource Consumption

Vegetable sellers were encouraged to reduce waste generation by properly handling vegetables and avoiding unnecessary disposal of usable products.

Discussions focused on translating awareness into measurable reductions in resource use.

6.3. Reuse – Extend Product Life Cycle

Students presented reuse-oriented solutions intended to extend product life cycles and minimize disposal. Stakeholders discussed mechanisms for practical application that:

Cardboard boxes and packaging materials from mobile shops can be reused for storage and transportation purposes instead of being discarded immediately.

6.4. Recycle – Recover Materials

Students introduced a recycling framework centered on waste segregation and material recovery. Stakeholders discussed operational requirements and implementation possibilities in such a way:

Mobile shopkeepers agreed to separate cardboard cartons, paper labels, and packaging materials so that recyclable items can be collected by waste collectors.

Discussions emphasized the importance of user-friendly segregation systems.

6.5. Rot – Compost Organic Waste

Since most of the waste at Old College Road consists of organic material, discussions were held with a nearby dairy farm owner. So, there is no need of rotting. The dairy farm owner agreed to collect vegetable waste regularly and use it as animal feed. Vegetable sellers also agreed to separate organic waste for easy collection.

7. Stakeholders Feedback and Negotiation Outcomes

Stakeholders appreciated the proposed waste management plan because it is simple, practical and does not require significant financial resources.

Key feedback included:

- Vegetable sellers agreed to separate organic waste.
- The dairy farm owner agreed to collect organic waste from the area.
- Mobile shopkeepers agreed to keep packaging waste separate.
- Community members supported efforts to improve cleanliness around Old College Road.



Based on stakeholder feedback, the following negotiated decisions were incorporated into the final plan:

- Organic waste will be collected by the dairy farm owner for animal feeding.
- Vegetable sellers will maintain separate collection points for organic waste.
- Packaging waste from mobile shops will be stored separately for disposal or recycling.
- Community cooperation will be encouraged to maintain cleanliness.

8. Final Agreed Implementation Plan

After negotiation, a consensus-based implementation plan was developed:

- Separation of organic waste at source.
- Regular collection of organic waste by the nearby dairy farm.
- Separate storage of cardboard and packaging waste by shopkeepers.
- Proper disposal and recycling of packaging materials.
- Community awareness regarding waste management and cleanliness.

9. Roles and Responsibilities

Stakeholder	Responsibility
Dairy Farm Owner	Collection of organic waste and use as animal feed.
Vegetable Sellers	Separation of organic waste.
Student Team	Awareness, monitoring and participation
Community Members	Household participation and compliance

Table 2: Stakeholders and their Agreed Responsibilities

10. Expected Outcomes

After negotiations with stakeholders, the outcomes are:

- Reduction in waste accumulation at Old College Road.
- The vegetable seller agreed to deliver the vegetable waste to the dairy farm owner every day at **8:00 to 8:30 PM**.
- Our group analyzed the current waste disposal practice and confirmed that this solution is practical and sustainable.
- The mobile phone shopkeepers also agreed to **collect and store their electronic waste in a designated area** within their shops until it is properly disposed of or recycled.
- Hence, the community participation increased in waste management activities.

11. Conclusion

The eco-diplomacy meetings successfully resulted in a practical waste management solution for Old College Road. Through cooperation between vegetable sellers, shopkeepers, community members and the nearby dairy farm owner, a simple and sustainable waste management plan was developed.

Hence, our waste audit project successfully transformed proposed theoretical waste management strategies into a collectively negotiated and practically feasible implementation plan.